Courtyard enclosure study · closing the conditioned house around the open courtyard · Taif / Palm House V7.0

Design review Y-01 (critical) · 30 Oct 2026 · the windows follow `docs/DESIGN_LANGUAGE.md` (black steel, divided lites)

DECIDED: option A (owner, 30 Oct 2026, applied in V7.0): the projection 1.55 m, B3 / R11 / C21–C23 at x 5.80, W1 / W2 / W3 as in § 2. The "now" column below is the V6.3 state (edge x 5.60), kept as the record of the comparison.

1 · What is open today

The courtyard (4.7 × 6.5 m on axes, the sky opening 3.35 × 6.5 m east of the projection) is outdoors, as the owner wants. But three edges of the **conditioned** house are open to its air, so the "closed, filtered house" (R-25) cannot be closed:

- **the gallery's edge** on x 5.6, y 6.5–19.25 (12.75 m): a railing on the upturned beam B3, over the sky (y 6.5–13) and over the double-height void (y 13–19.25);
- **the family living's edge** on y 6.5, x 4.25–8.95 (4.7 m): a railing on the edge beam B2;
- **the living's courtyard side above its glass**: the line y 13 between C22 and the kitchen wall is open from the glass head +2.90 to the roof.

2 · The proposal: one continuous line, three steel windows

The air / thermal line runs **on the structure that is already there**: along the gallery on the upstand of B3 (the glass sits on the upstand top at +3.82, 0.22 above the gallery floor: the splash and dust line of a courtyard), along the family living's edge on B2 from the floor, and on y 13 over the living's glass up to the roof edge beam R2. All three heads meet the one soffit line of the roof edge beams R1 / R2 / R11 at **+6.75**, so the courtyard reads as one room with one cornice.

Window	Line	Sill	Head	Lites and operation
W1 gallery edge	x 5.8 (A) / 5.6 (B), y 6.5–19.25	+3.82 (upstand)	+6.75	guarding transom at +4.70 (1.10 above the floor), laminated glass below it; above it lites ≈ 0.75 wide; one inward side-hung casement per bay (two bays, four casements) for calm nights and cleaning
W2 family living edge	y 6.5, x 4.25–8.95	+3.60 (floor)	+6.75	same transom at +4.70, laminated below; two inward casements; no door (the courtyard is reached from the ground floor)
W3 living, upper courtyard side	y 13, C22 → x 8.95	+2.90 (the glass head) or ±0.00 if the ground-floor glass is remade as one steel wall	+6.75	transoms at +2.90 and +3.82 (the gallery's sill line), lites ≈ 0.75 wide; two motorised bottom-hung tilt lites at +3.82–+5.40 for the night mode

Cleaning from inside. W1: both faces from the gallery: the inside directly, the outside through the open casements (every lite within an arm's reach of a casement, the glass height 2.93 m). W2: as W1 from the family living. W3: the inside from the living floor with a water-fed pole (top +6.75); the outside from the courtyard floor with the same pole (6.75 m) or from the gallery's casement at its west end. The courtyard floor's hose point (Y-04) serves the pole.
Frames. Black steel, thermally broken, divided lites, double low-e glass, dust seals, insect screens on the casements. Depth ≈ 0.06 m on the line; the stone column sleeves stop at the frame with a sealed joint (A) or stand free (B).

3 · The gallery: ≥ 1.20 m clear is the constraint

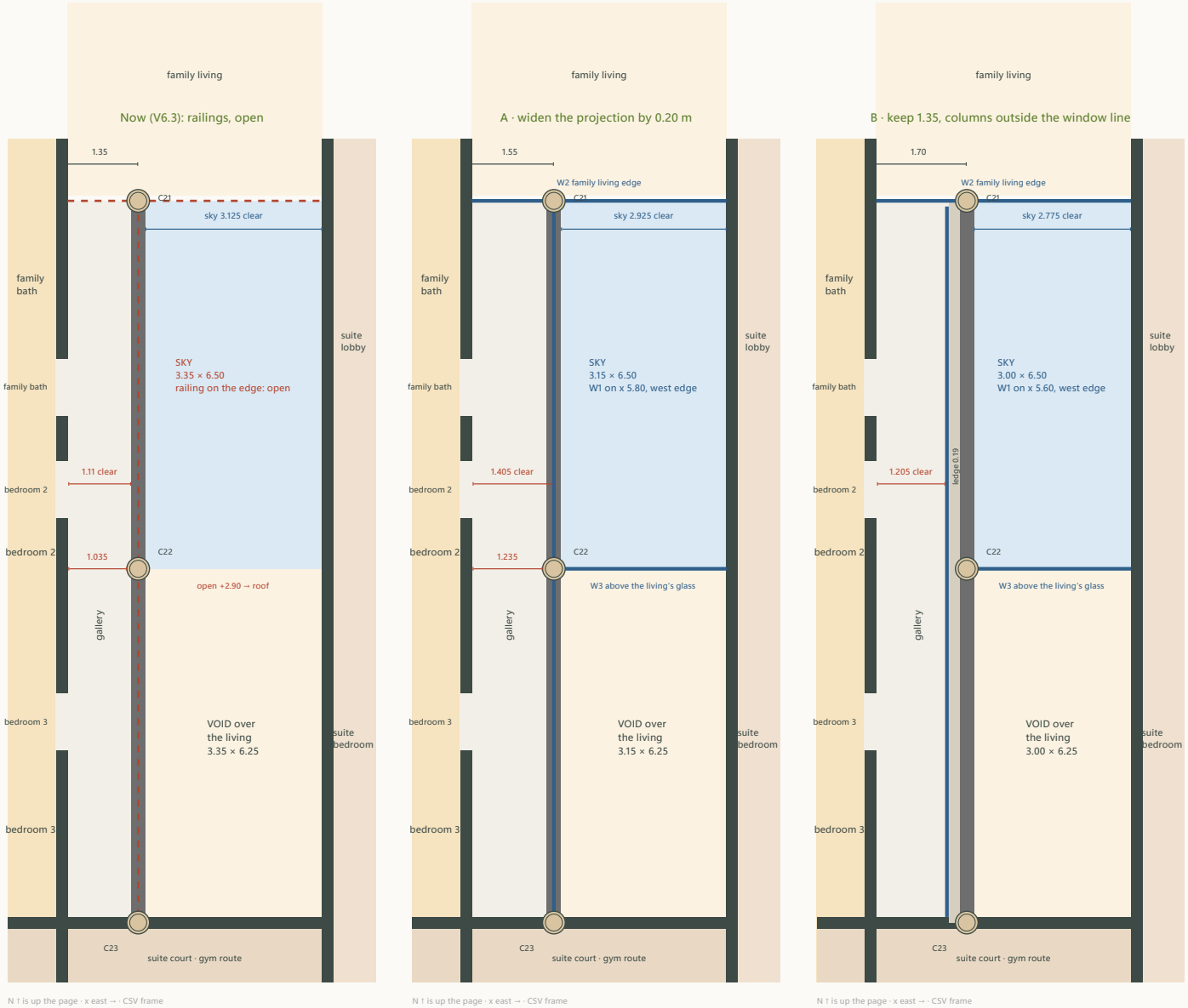
Today the gallery is 1.11 m clear between the rooms' finished wall face (x 4.365) and the upstand of B3, and 1.035 m at the three stone column sleeves (Ø0.4). A window on the line changes little by itself; the columns are the problem. Two ways to reach 1.20 m:

Item	Now (V6.3): railings, open	A · widen the projection by 0.20 m	B · keep 1.35, columns outside the window line
Projection on axes (x 4.25 → edge beam)	1.35 m (edge x 5.60)	1.55 m (edge x 5.80)	1.70 m (edge x 5.95)
Window line	—	x 5.8	x 5.6
Columns C21–C23 (stone sleeve Ø0.4)	x 5.6, in the line	x 5.8, in the line	x 5.95, 0.35 m outside the glass, free-standing
Gallery clear · minimum	**1.035 m** ✗ (< 1.20)	**1.235 m** ✓	**1.205 m** ✓
Gallery clear · at the columns	1.035 m	1.235 m	1.205 m
Gallery clear · at the window / railing	1.11 m	1.405 m	1.205 m
Gallery clear · at the bedroom doors	1.11 m	1.405 m	1.205 m
Sky opening, axes (clear · at the columns)	3.35 (3.125 · 3.05) m	3.15 (2.925 · 2.85) m	3.00 (2.775 · 2.7) m
Sky opening area, axes	21.78 m² (—)	20.47 m² (−6 %)	19.5 m² (−10.4 %)
Double-height void, axes (clear)	3.35 (3.125) m	3.15 (2.925) m	3.00 (2.775) m
Well: depth to the roof railing / width	2.48	2.63	2.77
Sky from a seated eye in the courtyard (share of the hemisphere)	7.7 %	7.2 %	6.8 %
Sky from the living's seated eye (light check point), through W3	3.2 %	3 % × 0.64 glass	2.8 % × 0.64 glass
New glass, about (W1 + W2 + W3)	—	60.16 m² (35.01 + 14.17 + 10.97)	63.34 m² (36.77 + 14.17 + 12.4)
Extra slab (first floor + roof strip)	—	2.55 + 1.3 m²	4.46 + 2.27 m²

Clear widths between finished faces: the rooms' wall 0.2 m + 0.02 plaster; the window frame 0.06 m centred on its line; the column sleeve Ø0.4 centred on the column. The doors (family bath y 9.3–10.3, bedroom 2 y 11.1–12.1, bedroom 3 y 15.2–16.2) are 0.7–2.2 m from the nearest column. "Sky from …": the solid angle of the sky opening (x from the beam's or the sleeve's face to the kitchen wall face, y 6.5–13, at +7.20) as a share of the hemisphere, the same measure for all three; × 0.64 for the glass of W3 (A and B alike), which today is open.

4 · Recommendation: A **A** keeps everything on one line: the beam, the columns, the window and the railing base are the same element, the gallery is 1.405 m clear and 1.235 m at the columns, the sky loses 6 % of its area and stays 3.15 m wide on axes. **B** reaches 1.20 m with no margin (1.205 m), narrows the sky more (3 m: the beam and the roof edge must follow the columns), and adds a 0.35 m ledge outside the glass that must be waterproofed, capped and kept clean; its one real gain, free-standing columns in front of a glass wall, is a stronger colonnade image. If the owner wants that image, B is buildable; otherwise A. In both, the two-storey columns, the upturned B3 and R11 stay as decided (X8); only their x changes. **Study checks passed** (7)

Plans of the gallery strip, first floor · now / A / B · the sky opening and the void · about 1:115 on A4



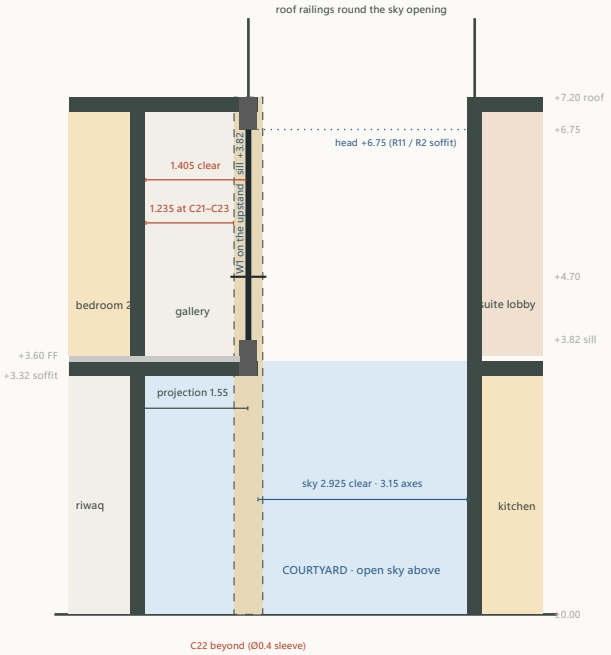
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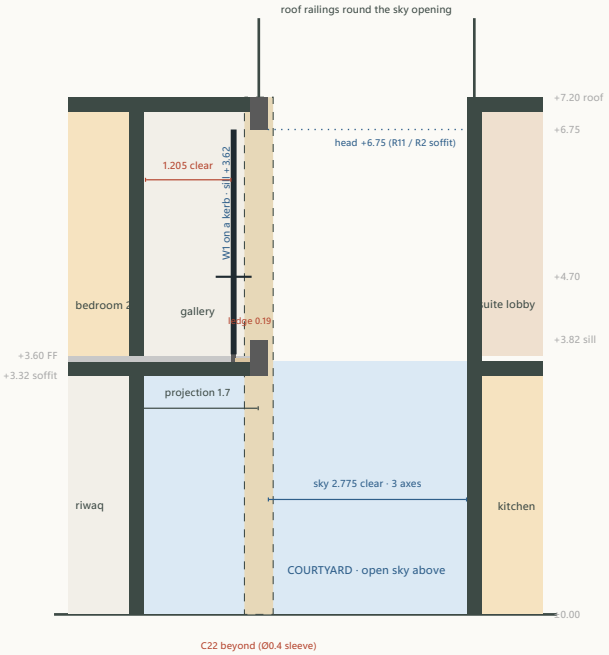
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Dark: walls and the edge beam B3 (0.25 wide, upturned 0.30 above the slab). Blue lines: the new steel windows W1 / W2 / W3; red dashed: the open railings of today. Columns: the stone sleeve Ø0.4 round the Ø0.3 core. In B the beam and the columns leave the window line and a 0.19 m ledge lies between the glass and the beam (capped in stone, falling to the courtyard). The gallery doors are the V6.3 1.00 m openings.

Sections through the courtyard (y ≈ 10, looking north) · A and B · about 1:105 on A4



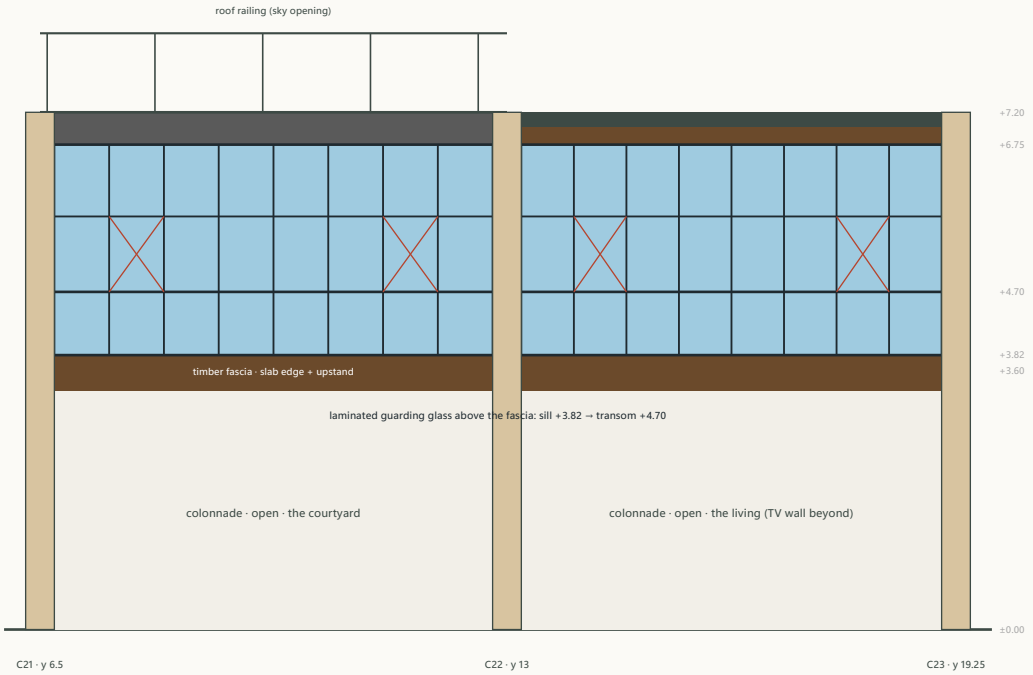
A · widen the projection by 0.20 m · section at y ≈ 10 looking north



B · keep 1.35, columns outside the window line · section at y ≈ 10 looking north

Dark: walls, slabs, the roof; grey: the upturned edge beam B3 (0.25 × 0.50 total, 0.30 above the slab) and the roof edge beam R11 above it; the dashed column is C22 (y 13.0) beyond the cut. W1 (heavy black line) sits on the upstand (A) or on a 0.10 m kerb inboard of it with a stone-capped ledge outside (B); its guarding transom at +4.70, its head at +6.75 under R11.

W1 · the gallery line seen from the courtyard and the void (looking west) · option A · 1:70 on A4



black steel, divided lites ≈ 0.75 wide · transoms at +4.70 (guarding) and +5.75 · red X = inward side-hung casements, two per bay (night mode, cleaning) · about 35.01 m² of glass · A · widen the projection by 0.20 m

Structure, daylight, cost, and what else changes

5 · Structure (studies/structure.typ)

Both options. W1 bears on the upstand of B3 (the window's dead load is small; the upstand becomes its sill and its anchor, 0.25 wide); W2 on B2 (floor-level sill, the frame anchored to the slab edge and R1); W3 spans +2.90 → +6.75 between C22 and the kitchen wall, held top and bottom (R2 above; a steel transom at +3.82 tied to C22 and the wall takes the wind of the 3.85 m height; the courtyard well has little wind). The head line is the soffit of R1 / R2 / R11 at +6.75 over the void (y 13–19.25) a 0.25 m fascia keeps the same line under the roof slab. Nothing new for the engineer except the window anchors and, in B, the ledge.

- **A:** B3, R11 and C21–C23 move 0.20 m east to x 5.8 the projection slab bay becomes 1.55 m (slabs-ff row "the projection"), the roof strip over the gallery 1.55 m (slabs-roof); C22 stays the pier in the living's courtyard glass (now at x 5.8), C21 in the women's majlis glass, C23 in the living's south wall; the spans (6.5 / 6.25) are unchanged, R2's west end lands on C22 at x 5.8. The sky opening is 3.15 × 6.5 m.
- **B:** B3, R11 and C21–C23 move 0.35 m east to x 5.95 the slab bay becomes 1.7 m with the window line 0.35 m inboard of the beam on a 0.10 m kerb; the ledge between the kerb and the upstand is waterproofed and capped (stone, falling out). The columns stand free in the courtyard and in the void: C22 is a free column in the living's courtyard glass line, 0.35 m in front of the upper window W3 (W3 then runs behind it, from the W1 line at x 5.6 to the kitchen wall, 3.25 m). Alternative B2 (not recommended): B3 stays at x 5.6 and reaches the columns at x 5.95 on short corbels; two lines of concrete for one edge.
- The courtyard colonnade below: in A the columns move with the slab edge (the covered strip 1.55 m on axes); in B the columns stand 0.35 m further into the courtyard than the glass above, and the covered strip is 1.7 m.

6 · Daylight (simple, as in the light check)

The courtyard well is 8.3 m deep to the roof railing. Today it is 3.35 m wide (ratio 2.48); A 3.15 m (2.63); B 3 m (2.77). The sky seen from a seated eye in the middle of the courtyard floor is 7.7 % of the hemisphere today, 7.2 % in A and 6.8 % in B. From the living's seated eye (the light check point, through the y 13 line) it is 3.2 % today; A 3 % and B 2.8 % **through the new glass** (× 0.64 for the glass and its frames, A and B alike: the living's upper light comes through W3 in both). In short: A costs the living about 6.2 % of its sky view by geometry and ≈ 36.3 % by the glass; B 10.9 % and the same glass. The clerestory light (sections.typ: winter noon sun on the clerestories, the suite court) is untouched by either. The courtyard's own light drops a little in both (the well narrows by 0.2 / 0.35 m); the pergola (character study, over y 6.5–9.5) takes more than either option and should be reconsidered when the enclosure is decided.

7 · Cost and buildability

- **Glass:** about 60.16 m² (A) / 63.34 m² (B) of thermally broken black steel windows with divided lites and double low-e glass, laminated below the guarding transom; six casements and two motorised tilt lites. If the ground-floor living glass (y 13) is remade as part of one steel wall ±0.00 → +6.75, add ≈ 8.26 m². This is the main cost and is the same in A and B.
- **Concrete:** A adds 2.55 m² of first-floor slab and 1.3 m² of roof strip; B adds 4.46 + 2.27 m² plus the kerb, the ledge waterproofing and 12.75 m of stone capping. Both are small.
- **Build:** A is one line: the formwork of B3 is also the sill, the columns meet the frames with one sealed joint each (six joints), and the soffit, the fascia and the railing base are as decided. B has two lines to coordinate (beam and kerb), a ledge that collects dust and water and is reached only through the casements, and columns that must be cast free-standing to the roof with the frames passing behind them (the frames are then continuous, 12.55 m, with no column joints).
- **Guarding:** the window itself guards the gallery and the family living (laminated glass to +4.70, the transom at 1.10 above the floor): the bar railing of the design language is then the stair's, not the gallery's. If the owner wants a bar railing on the gallery as well, it stands inside the glass and takes ≈ 0.05 m of the clear width (A still ≥ 1.20; B does not).

8 · Knock-on effects

- **The malqaf and the courtyard air.** The courtyard stays outside and the house is sealed round it, so the malqaf's grille at the foot of the kitchen wall serves only the courtyard: in the closed (dust) mode nothing of it reaches the rooms, and the courtyard well behaves as a chimney, cool air at the bottom, warm air out at the top. In the night mode the casements of W1 / W2 and the tilt lites of W3 open and the courtyard's tempered air enters the gallery, the family living and the living: the malqaf keeps a use, but only with the windows open (insect screens, a damper interlock with the ERV). The review's E-01 stands: decide the malqaf's role with this item (feature only; the ERV's intake; or an atrium). The sunken seating, the fountain and the pergola stay outdoor elements.
- **The clerestory windows** (y 19.25, +5.00 … +6.00, onto the suite court) are not on the courtyard and do not change; after the enclosure they and W3's tilt lites are the living's only openings: keep them fixed in the closed mode or make two of them operable with the night mode, and plan their cleaning from the suite court.
- **The AC and ERV:** the living, the gallery and the family living become conditioned volumes (the double-height living 4.7 × 6.25 × 6.75 m): the MEP engineer sizes for them; the glass faces a shaded well, so the solar gain is modest.
- **The character study** (3d/character.yaml): the colonnade's arches are gone (V6.3); the gallery railing variants (parapet / glass) are replaced by W1 on the upstand; the lanterns hang under the soffit as before; the roof railings round the sky opening stay.
- **Other review items** that this decides or touches: F-01 (gallery width: solved by A or B), F-06 (family living edge: W2), G-02 (the living's light: W3's glass, § 6), Y-04 (cleaning: the hose point and the pole), E-01 (malqaf).

9 · Decision needed

A or B (or the atrium, which this study does not draw). On the decision the next step changes `data/rooms_ff.csv` (the gallery polygon), `studies/ff_family.typ` (proj-x, columns), `studies/structure.typ` (C21–C23, B3, R11, the slab bays), `presentation.typ` (ff-walls: windows instead of rails), `3d/open_edges.csv` and `3d/params.yaml` (W1–W3 as windows), the light check and the character study.